

this collector current is a tedious exercise, as the value varies with applied voltage, resistors, MOSFET parameters, load current and so on. The easiest option is to simulate the circuit.

Simulation

Simulation software is recommended to simulate and analyse the circuit before implementation. Here, we used LT SPICE, a free simulation software. Simulations are performed by applying a triangular wave of amplitude 0V-5V, 1Hz at anode side and fixed 2.5V at cathode side.

As explained earlier, once MOSFET is connected, waveform at collector of Q2B will get modified depending on load current. To avoid the effect of voltage sources on load current, two ideal diodes (Q1 and Q3) are connected back-to-back with a variable load connected between these diodes, as shown in Fig. 2. Waveforms captured for left-side ideal diode (Q1) are shown in Figs 3 and 4.

Results in Figs 3 and 4 clearly show that voltage across cathode side transistor Q2B depends on load current (I_L).

Dependence on V_{GS} may be observed by using a different MOSFET in simulation or by editing SPICE model. There is a small voltage drop on cathode voltage due to resistive voltage drop of MOSFET, which is not quite evident in the waveforms as resistance of the selected MOSFET is quite low.

Design considerations

1. As mentioned earlier, the major parameter to be considered while selecting the BJTs is difference in V_{BE} voltages. Mismatch in V_{BE} voltages of transistors can cause the MOSFET to be undesirably on or off for small

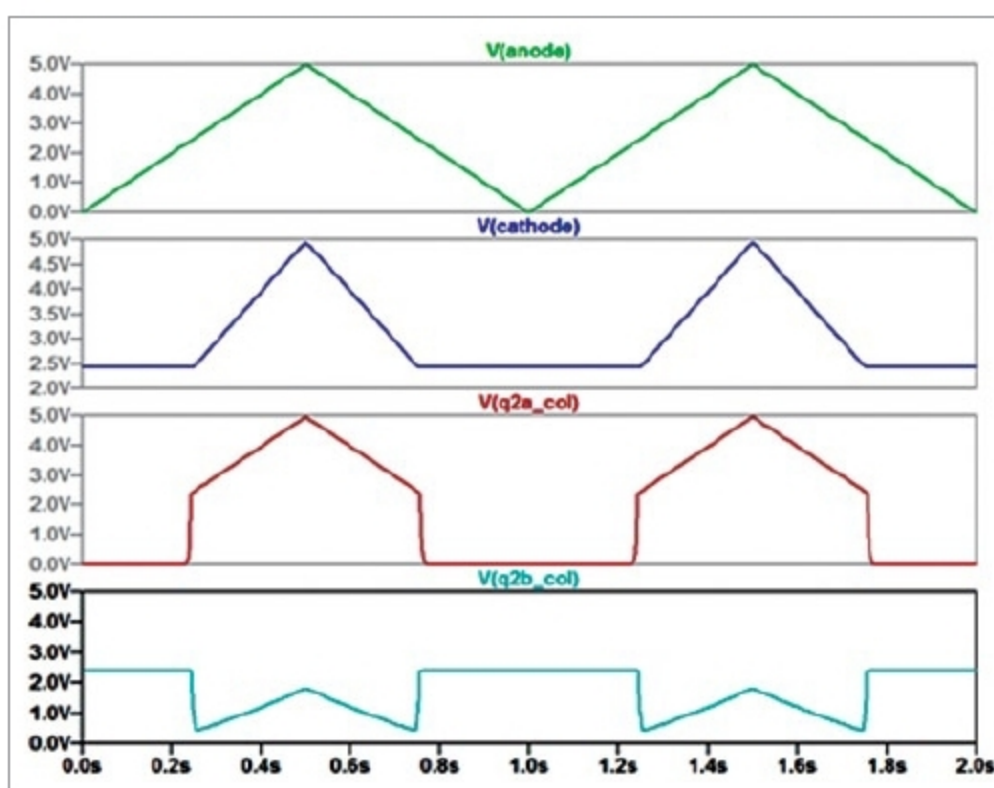


Fig. 3: Simulation waveforms, load current=500mA

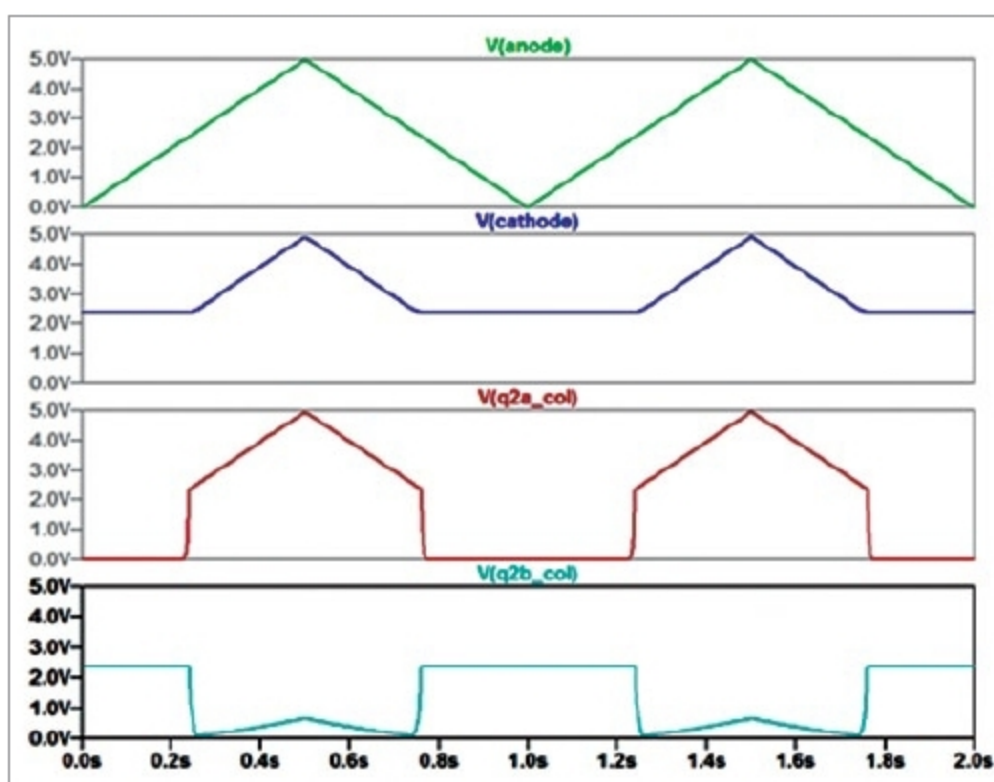


Fig. 4: Simulation waveforms, load current=1A

differences in voltages on either side.

If voltage difference between anode and cathode sides is large, any suitable transistors can be used. But this may not be the case for majority of the designs calling for the need of matched pair transistors.

2. The next parameter to be considered is saturation current ($I_{C(Sat)}$). Since saturation current of the transistor is leakage current in the circuit,

this should be kept as low as required.

Collector-emitter saturation voltage ($V_{CE(sat)}$) of the BJT is another parameter to be considered, which will affect the selection of MOSFET. If there is any chance of negative voltage stress on the BJTs, emitter-base breakdown voltage (BV_{EBO}) might also need to be considered.

3. Selection of MOSFET largely depends on application. Major parameters to be considered are drain-source breakdown voltage (BV_{DSS}), static drain-source on-resistance ($R_{DS(ON)}$), maximum continuous drain current (I_D) and gate threshold voltage.

Drain-source voltage should be selected as per maximum possible voltage difference between anode and cathode sides. On-state resistance of MOSFET will cause voltage drop across the FET, which would become the effective forward drop of the diode.

$R_{DS(ON)}$ shall be kept as low as possible to keep voltage drop at minimum. Gate threshold voltage should be selected so that absolute value of minimum $V_{GS(th)}$ is more than $V_{CE(sat)}$ of transistor and maximum is less than minimum voltage applied to the MOSFET ($V_{CE(sat)} < |V_{GS(th)}| < V_{IN(Min)}$).

4. Design of the circuit is quite simple, as it only requires the use of both transistors as switches. Collector resistances and base resistance need to be calculated properly to ensure the transistor will be in saturation mode for the expected voltage range of operation, yet saturation current should be as low as acceptable. **EFY**

EFY Note

The source code of this project is available for free download at source.efymag.com

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